

CBPROG 2 MACHINE PROJECT

“Embodied Emissions Calculator”

By Antonio Gabriel M. Imatong and Andrei John Sumilang

I. Introduction

With the rising concern for climate change and global warming each year, big steps must be taken to mitigate the human activities that cause it. With this in mind this was the inspiration for our machine project. Our machine project covers SDG #9 and SDG #11 which are industry innovation and infrastructure and sustainable cities and communities. This machine project will be a program that construction companies can use to see if their project plans are within a safe amount of embodied emissions (emissions caused by building infrastructure projects) are within a safe range to be on track to mitigate the effects of climate change specifically all projects must reduce by [40% by 2030](#).

The data for the materials and their corresponding embodied emission value in this database is from the [ICE Database](#) by Circular Ecology which compiled all of the different embodied carbon values from multiple studies and a company's EPD. Because of this there are some limitations to what products can be put into the database of our machine project. For the sake of simplicity and limitations for the project we provided density constant per category of material.

II. Process

I User-Input

- A. Username
- B. Password
- C. Project Info

II. Project Info (Company/Residential)

- A. Section

1. Dimensions
2. Product Code

III. Process Information

A. Sum of Actual EE

1. Use product code to locate EE amount of given product
2. Compute using the given EE amount of specified product
3. Get sum according to the given sections of the project
4. EE Sum

B. Sum to compute for ceiling

1. Identify the material type of the given section
2. Use EE of the highest amount in each material type
3. Get sum according to the given sections of the project
4. Subtract 40% from the total (60% of the sum)
5. Ceiling

IV. Approval

If EE Sum $\leq$ Ceiling
 Project Approved
 Else
 Project Rejected

V. User Access

- Application Admin
 - Access records from different companies
 - Add new products to catalogue
- Project Manager
 - Create Projects
- User
 - View Projects

III. Additional-Information

The project will have a database for the following:

- Brands/type of all construction materials
- Project history of all registered companies
- Registered Users
- The emissions of each individual material

The project will compute for the following:

- The total embodied emissions
- How much emission per material has given the project dimensions
- Converting volume of section to kg CO₂e
- Computation for total amount of emissions
- Computation for ceiling

TESTING

encrypt.c						
Function name	#	Description	Input	Expected output	Actual output	P or F
encrypt	1	Encrypting password	“admin”	“NKhVU”	“NKhVU”	P
decrypt	2	Decrypting password	“NKhVU”	“admin”	“admin”	P
is_equal	3	Valid password	“NKhVU”, “admin”	1 (true)	1	P
		Invalid password	“NKhVU”, “user”	0 (false)	0	P
is_encryptable	4	Within encryptable range	“CaesarCipherC”	1	1	P
		Not within encryptable range (contains space)	“Hello, World”	0	0	P

io_util.c						
Function name	#	Description	Input	Expected output	Actual output	P or F
io_get_int	1	Valid int	1	1	1	P
		Invalid int	abc	Retry input	Retry input	P
io_get_uint	2	Valid unsigned integer	125	125	125	P
		Invalid unsigned integer	-123	Retry input	Retry input	P
io_get_float	3	Valid float	143.23	143.230000	143.230000	P
		Invalid float	ghi	Retry input	Retry input	P
io_get_string	4	Within maximum buffer size	“abc”, Size = 4	abc\n	abc\n	P
		Beyond maximum buffer size	“abcdef” Size = 5	Retry input	Retry input	P

menu.h						
Function name	#	Description	Input	Expected output	Actual output	P or F
run_menu_internal	1	Input not within the menu	MenuItems: List, Add, Delete Selection: 4	-1	-1	P
		Input within the menu	MenuItems: List, Add, Delete Selection: 3	2	2	P

project.c						
Function name	#	Description	Input	Expected output	Actual output	P or F
	1					P
						P
	2					P
						P
	3					P
						P
	4					P
						P
	6					P
						P
	7					P
						P
	8					P
						P

	9					P
						P
	10					P
						P
	11					P
						P
	12					P
						P
	13					P
						P
	14					P
						P
	15					
	16					P
						P
calculate_project_ee_ceiling	17		material-list			P
						P

string_util.c						
Function name	#	Description	Input	Expected output	Actual output	P or F
str_equal	1	Equal string	a = “Hello” b = “Hello”	1	1	P
		Not equal string	a = “Hello” b = “Hi”	0	0	P

str_split	2	Three strings with a space as delimiter	“Hello World Test”	3	3	P
		Two strings with ‘,’ as limiter	“Hello,World”	2	2	P
substring	3	Assign Hello to array index 0	dest [0], “Hello World”	“Hello”	“Hello”	P

user.h						
Function name	#	Description	Input	Expected output	Actual output	P or F
read_cred_element	1	File is Not Null		1	1	P
		File is Null		0	0	P
write_cred_element	2					P
						P
load_creds_from_file	3					P
						P
save_creds_to_file	4					P
						P
read_user_element	5	File is Not Null		1	1	P
		File is Null		0	0	P
write_user_element	6					P
						P
load_users_from_file	7					P
						P
save_users_to_file	8					P
						P
get_cred_by_id	9					P

						P
get_user_by_id	10					P
						P
register_user	11					P
						P
login_user	12					P
						P
verify_security_answer	13					P
						P
change_password	14					P
						P
update_credential	15					P
						P
update_user	16					P
						P
delete_user	17					P
						P
user_comparator	18					P
						P
has_permission	19					P
						P

WELCOME TO THE EMBODIED EMISSION CALCULATOR
A Lasallian Initiative for a Greener Future

[SDG 9: Industry & Innovation] [SDG 11: Sustainable Cities]

Inspired by the UN Sustainable Development Goals, this program empowers the construction industry to track carbon footprints.

TARGET: 40% Carbon Reduction by 2030

> Live the mission. Build the future. Animo La Salle!

[Info] Press [ENTER] to continue...

Authentication

[0] Back/Exit

[1] Login

[2] Forgot Password

[Input] Selection: 1

[Input] Enter User ID: 1

[Input] Enter Password: tes

[Error] Login Failed: Invalid Credentials.

Authentication

[0] Back/Exit

[1] Login

[2] Forgot Password

[Input] Selection: 1

[Input] Enter User ID: 1

[Input] Enter Password: admin

[Success] Login Successfull. Welcome Back!

[Info] Press [ENTER] to continue...

Main Menu

[0] Back/Exit

[1] Materials

[2] Projects

[3] Accounts

[Input] Selection: 2

Project Management

[0] Back/Exit

[1] List Projects

[2] View Project

[3] Add Project

[4] Delete Project

[Input] Selection: 1

Project List

ID	Project Name	Sections	Embodied Emission	Status
123	APPROVED PROJECT	4	7569486.00	Safe
124	UNAPPROVED PROJECT	4	14027044.00	Danger
125	test	1	84686.00	Safe

- [0] Back/Exit
- [1] List Projects
- [2] View Project
- [3] Add Project
- [4] Delete Project

[Input] Selection: 3

Create New Project

[Input] Project ID: 222

[Input] Project Name: New

Total Embodied Emission: 0.00/0.00 [Normal]

PROJECT: New (ID: 222)

ID	Section	Material	Vol (m3)	Embodied Em.
-----	-----	-----	-----	-----

TOTAL EMBODIED CARBON: 0.00 kg CO2e

END OF PROJECT REPORT

Project Detail View

- [0] Back/Exit
- [1] Add Section
- [2] Delete Section

[Input] Selection: 1

[Input] Section Name: Wall

Material Catalog

ID	Company	Category	Type	Product	Carbon	Density
1	Holcim Philippines Inc.	Cement	Fly Ash Cement	Holcim EcoPlanet	0.59	2400.00
2	Holcim Philippines Inc.	Cement	Port-Poz. Cement	Holcim Solid	0.74	2400.00
3	Republic Cement	Cement	Type II Portland - Lime	Rapid Set PLUS	0.61	2400.00
4	Lemery Corp	Cement	Blast Furnace	eco TIBAY Slag	0.34	2400.00
5	Dangote Cement	Cement	Portland - Lime	Dangote 3x 42.5N	0.74	2400.00
6	Vicat	Cement	Pozzolanic Cement	Vicat DECA	0.46	2400.00
7	Anhui Conch Cement Co.	Cement	Port - Slag	Conch Brand	0.75	2400.00
8	Philippines Wood	Wood	Hardwood	N/A	0.30	644.00
9	Philippines Wood	Wood	Softwood	N/A	0.26	644.00
10	Philippines Wood	Wood	Plywood	N/A	0.68	644.00
11	MATIMCO	Wood	Laminated Veneer Lumber	N/A	0.39	644.00
12	Filtra Timber	Wood	Softwood	N/A	0.26	644.00
13	Filtra Timber	Wood	Hardwood	N/A	0.30	644.00

14	Falcon Timber	Wood	Chipboard	
N/A	0.45	644.00		
15	Ten Build PH	Wood	Fibreboard	N/A
0.71	644.00			
16	Saint Gobain Glass	Glass	Laminated Toughened Glass	
ORAE low carbon glass	0.27	2500.00		
17	Saint Gobain Glass	Glass	Triple Glazed	
Clima Top Triple Glazed	1.75	2500.00		
18	Saint Gobain Glass	Glass	Double Glazed	
Clima Top Double Glazed	1.63	2500.00		
19	Lamilux	Glass	Toughened Glass	
Glass Roof Passive Standard	3.10	2500.00		
20	Lamilux	Glass	Multilayer Safety Glass	
Glass Roof Fire Resistance	2.08	2500.00		
21	Pacific Glass Corporation	Glass	Single Glass Pane	
Tinted Float Glass	1.44	2500.00		
22	Carlton Glass Enterprise	Glass	Double Glazing	
DGU Double Glaze	1.63	2500.00		
23	Viridian Glass	Glass	Double Glazing	
ThermoTech	2.08	2500.00		
24	Viridian Glass	Glass	Toughened Glass	
Intruder Guard	1.66	2500.00		
25	Accutech Steel	Steel	Electrogalvanized	
N/A	2.71	7840.00		
26	Shoji Steel Inc.	Steel	Electrogalvanized	N/A
2.71	7840.00			
27	Linton Inc.	Steel	Cold Rolled Coil	N/A
2.47	7840.00			
28	Linton Inc.	Steel	Galvanized	N/A
1.93	7840.00			
29	Steel Asia	Steel	Rebar	N/A
1.72	7840.00			
30	Metal Exponents Inc	Steel	Hot Rolled Coil	
N/A	2.38	7840.00		

31	Metal Exponents Inc	Steel	Cold Rolled Coil	
N/A	2.47	7840.00		
32	Dek Metal	Steel	Organic Coated	N/A
2.88	7840.00			

[Input] Select Material ID: 4

[Input] Length (m): 10

[Input] Width (m): 10

[Input] Height (m): 5

[Success] Section added.

Total Embodied Emission: 410400.00/543600.00 [Normal]

PROJECT: New (ID: 222)

ID	Section	Material	Vol (m3)	Embodied Em.
1	Wall	eco TIBAY Slag	500.00	410400.00

TOTAL EMBODIED CARBON: 410400.00 kg CO2e

END OF PROJECT REPORT